

Artificial Intelligence

Orbital Compute: Starcloud - Day 3 Takeaways from Citi's Global TMT Conference

CITI'S TAKE

We hosted Philip Johnston, the CEO of Starcloud, for a fireside chat on September 10, 2026, at our Global TMT Conference in NYC. Starcloud is a space infrastructure company developing orbital data centers, which it believes could help meet rising AI compute demand while avoiding some of the power, land, and permitting constraints associated with terrestrial data centers. The discussion focused on launch capacity as the key constraint on commercial deployment, government inference workloads as an initial high-paying market, and the technical milestones needed to make orbital compute competitive with terrestrial infrastructure. Key takeaways follow.

Johnston thinks lower launch costs could make orbital compute competitive with new terrestrial capacity in the early 2030s. He estimates the model becomes viable at roughly \$800/kg, compared with approximately \$5K/kg today. A Starship could carry roughly 50 of their satellites, collectively representing 50MW of capacity. If launch costs and frequency improve, Johnston expects annual orbital data-center capex could reach ~\$1T within a decade. He believes orbital systems could eventually account for more than 50% of new compute capacity while remaining below 5% of the installed base because terrestrial infrastructure will turn over slowly.

Government inference provides an initial market while Starcloud waits for commercial launch capacity. Starcloud expects government customers to represent nearly 100% of their workloads over the next three years because processing satellite data in orbit can bypass constrained downlinks. Johnston noted that satellites may currently transmit only ~1% of collected data over three days, while in-orbit inference could process the full dataset and return only the information needed. He expects orbital compute to remain focused on inference because optical links cannot support the GPU clustering required for frontier-model training. Starcloud expects commercial Starship payloads to begin ramping in mid-2028, consistent with SpaceX's expected delivery.

Starcloud has demonstrated orbital GPU performance and is now progressing toward in-orbit validation of the cooling, radiation-mitigation, and manufacturing capabilities needed to reach commercial scale. Starcloud-1 carried an NVIDIA H100, trained NanoGPT, and ran Google's Gemma model in orbit. For Starcloud-2, the two main technical challenges are dissipating heat in a vacuum without heavy radiators and protecting GPUs from radiation. Johnston believes Starcloud has developed solutions to both engineering hurdles, although larger-scale validation remains ahead. Starcloud plans at least three launches in 2027 to

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test these technologies on larger systems before advancing to Starcloud-3, which will string together multiple Starcloud-2 spacecraft into a Starship-scale deployment. The company is working with NVIDIA on space-optimized hardware and has applied to operate up to 88K satellites, with Johnston targeting production of roughly 100 satellites per week once launch capacity scales. He thinks achieving that scale will depend primarily on repeatable manufacturing and reliable, lower-cost launch capacity.

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